

ELECTRIC AND DIESEL TRAINS



In 1964 the Japanese opened the first high-speed electric railroad. These “bullet” passenger trains reached speeds of 130 miles (210 kilometers) an hour, a world record at that time. But railroad companies earn most of their money by moving goods, called freight. Freight trains are often pulled by diesel locomotives, and they keep a lot of traffic off the roads. The world’s longest freight train had 682 cars and was more than four miles (seven kilometers) long!

Pantographs pick up electricity from overhead wires to power the train.



Diesel power

Many passengers and goods still travel by cheaper, diesel-powered trains, such as this long Canadian railroad convoy.

On electric trains, the section that pulls the coach cars is called a power car.



The French electric high-speed train is called the TGV. On its regular route it has a top speed of 190 mph (300 kmh).



Around the bend

Trains have to slow down to go around bends safely. Some trains can tilt inward on curves, just like a motorcycle leaning into a bend. This means they don’t have to slow down so much.

The Inter-City Express, or ICE, is an electric train from Germany.

High-speed train routes are expensive to set up because they need a special track that has gentle curves.